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Task 3: Functional Requirements

Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC001	Successful login with valid credentials	Username = admin, Password = 1234	Login Successful and Dashboard opens	Login Successful and Dashboard opens	PASS
TC002	Login with invalid username	Username = Kamogelo, Password = 1234	Error message: Invalid Credentials	Error message: Invalid Credentials	PASS
TC003	Login with Invalid password	Username = admin, Password = 4321	Error Message: Invalid Credentials	Error message: Invalid Credentials	PASS
TC004	Login with empty fields	Username = , Password =	Error message: Fields cannot be empty.	Error message: Invalid Credentials	PASS
TC005	Student Registration valid details	student ID =20240731 + Full Name=Kamogelo Phaladi + age = 21 + email= 20240731@gmail.com + select programme = Software Engineering	Student added successfully and summary display	Student added successfully and summary display	PASS
TC006	Register student with empty fields	student ID = + Full Name= + age = + email= + select programme =	Error message: Fields should not be empty	Fields should not be empty	PASS
TC007	Course Enroll student in a new course(valid)	Student ID=202470731+ Student name= Kamogelo + Selected course =Software testing + Select semester= semester 1	Success message + Course added to student	Success message + Course added to student	PASS
TC008	Enroll student with empty fields	Student ID= + Student name= + Selected course = + Select semester=	Error message: Fields should not be empty	Fields should not be empty	PASS
TC009	Marks capture, Valid numeric input	Student ID=20240731 + Student name= Kamogelo Subject1=88, Subject2=82, Subject3=90	Average calculated correctly (86.67)	86.66666667	PASS
TC010	Attempt to capture empty fields	Student ID= + Student name= + Subject1= , Subject2= , Subject3=	Error message: Fields should not be empty	Fields should not be empty	PASS

TC011	Attempt to capture non numeric input	Enter Student details + Subject1=pass, Subject2=fail, Subject3=pass	Validation error / Error message	Please enter valid number marks	PASS
TC012	Testing Maximum Average calculation	Enter Student details + Subject1=100, Subject2=100, Subject3=100	Average = 100.00	100.00	PASS
TC013	Testing Minimum Average calculation	Enter Student details + Subject1=0, Subject2=0, Subject3=0	Average = 0.00	0.00	PASS
TC014	Attempt to generate report with empty fields	Report type = + Student ID filter=	Error message: Fields should not be empty	Fields should not be empty	PASS
TC015	Performance testing report display	Generate any report	Report loads within acceptable time (under 2 seconds)	Report loads within 0.7 seconds	PASS
TC016	Logging out	Click logout on dashboard	Directed back to Login	Directed back to login	PASS

Task 4: Non-functional testing

I performed non-functional testing on the ULMS application focusing on the following areas:

4.1 Performance Testing

Method:

Timed the report generation process using the built-in Stopwatch and manual observation before and after optimization.

Test Details:

- Generated Student Summary, Marks and Enrolment Reports multiple times.
- Measured loading time.

Results:

Before optimization: Report generation took approximately 4-5 seconds (due to `Thread.Sleep(4000)`).

After optimization: Report generation now takes under 1 second.

Findings:

Performance has improved significantly. The system now meets acceptable response time standards for a desktop application.

4.2 Usability Testing

Umoja Learning Management System - Dashboard

ULMS Dashboard

Logout

Student Registration

Course Enrolment

Marks Capture

Reports

Umoja Learning Management System - Student Registration

Student ID

Full Name

Email

Age

Programme

Save Student

Clear

Back

Umoja Learning Management System - Course Enrolment

Student ID

Student Name

Course

Semester

Enroll Student

Clear

Back

Umoja Learning Management System - Marks Capture

Student ID

Student Name

Subject 1 Mark

Subject 2 Mark

Subject 3 Mark

Calculate Results

Clear

Back

Umoja Learning Management System - Reports

Report Type

▼

Student ID Filter

Report Output

Generate Report

Clear

Back

Method:

Evaluated the user interface and navigation flow, evaluating the ease of use consistency and feedback.

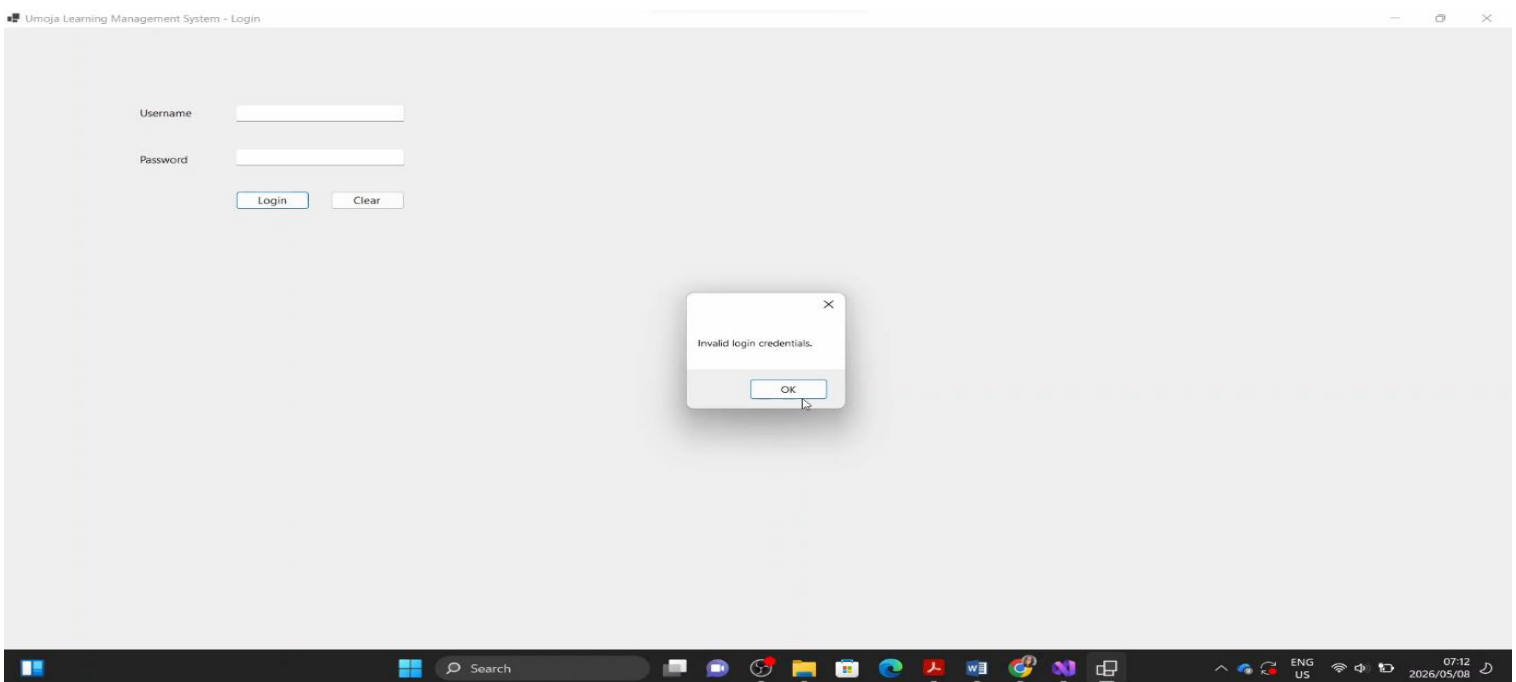
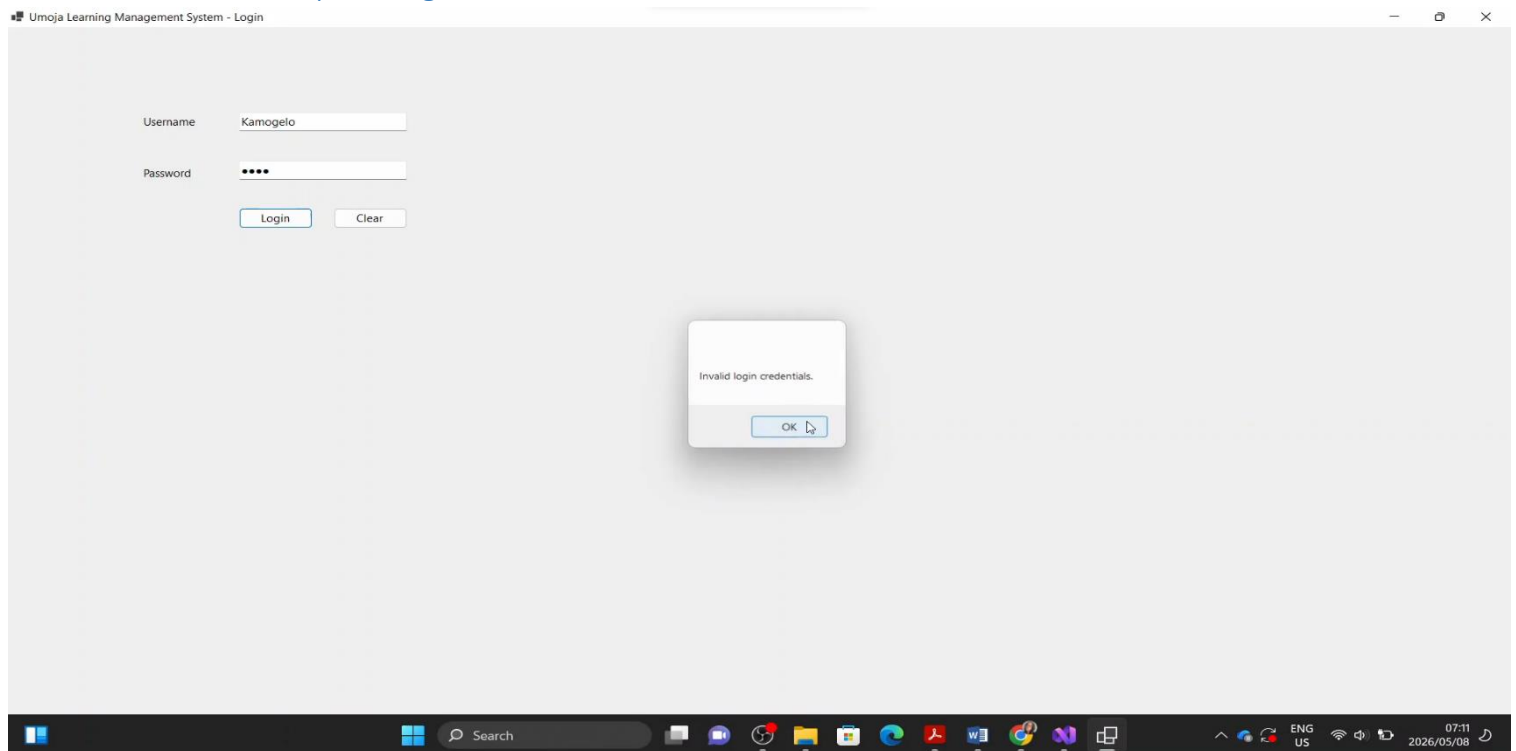
Results:

Aspect	Rating (1-5)	Comments
Navigation	5	Simple and intuitive dashboard
Input Fields	4	Clear labels good layout
Error Messages	5	Clear and helpful after fixes
Overall User Experience	4	User-friendly for a desktop LMS

Findings:

The interface is clean and easy to navigate.

4.3 Security Testing



Method:

Tested authentication and basic input validation vulnerabilities.

Test Cases Performed:

Multiple invalid login attempts

Empty field test

Results:

Login now properly validates both username and password after fix.

Basic input validation prevents most invalid entries.

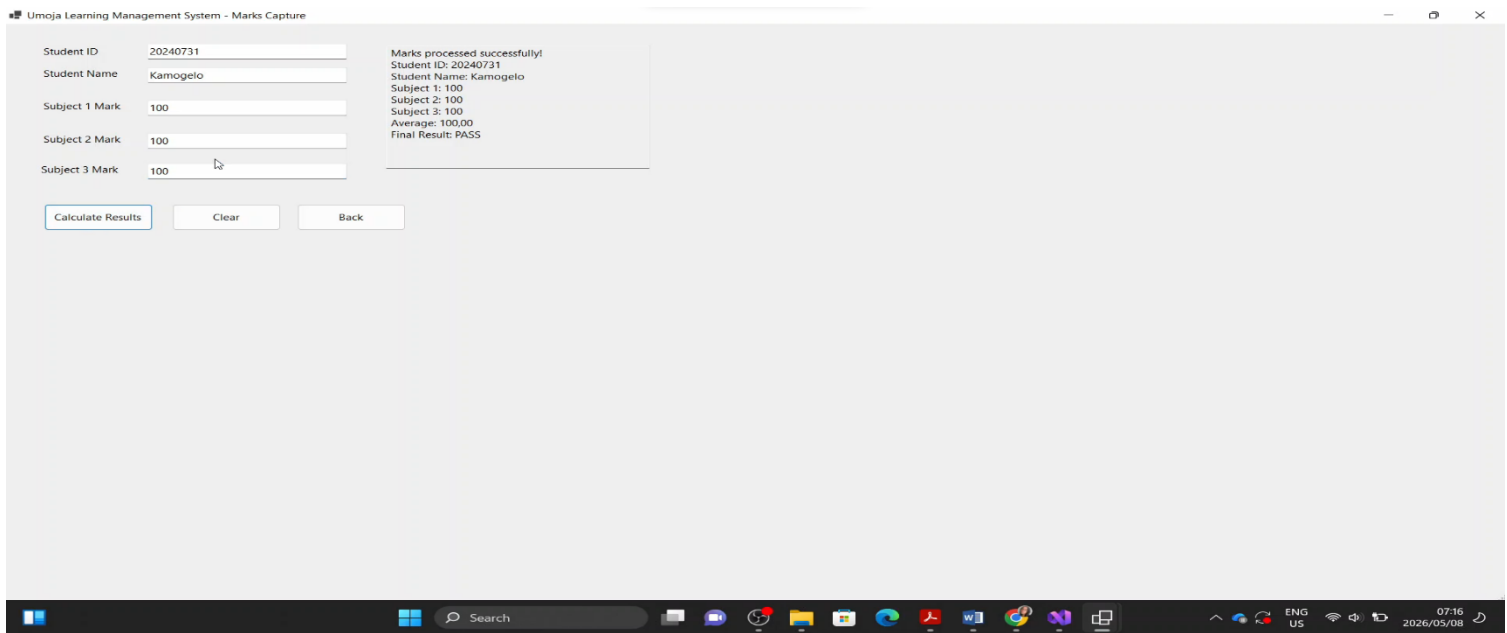
Findings:

Security has improved after fixing login logic. However, the application would benefit from proper authentication hashing in a real environment.

4.4 Reliability Testing

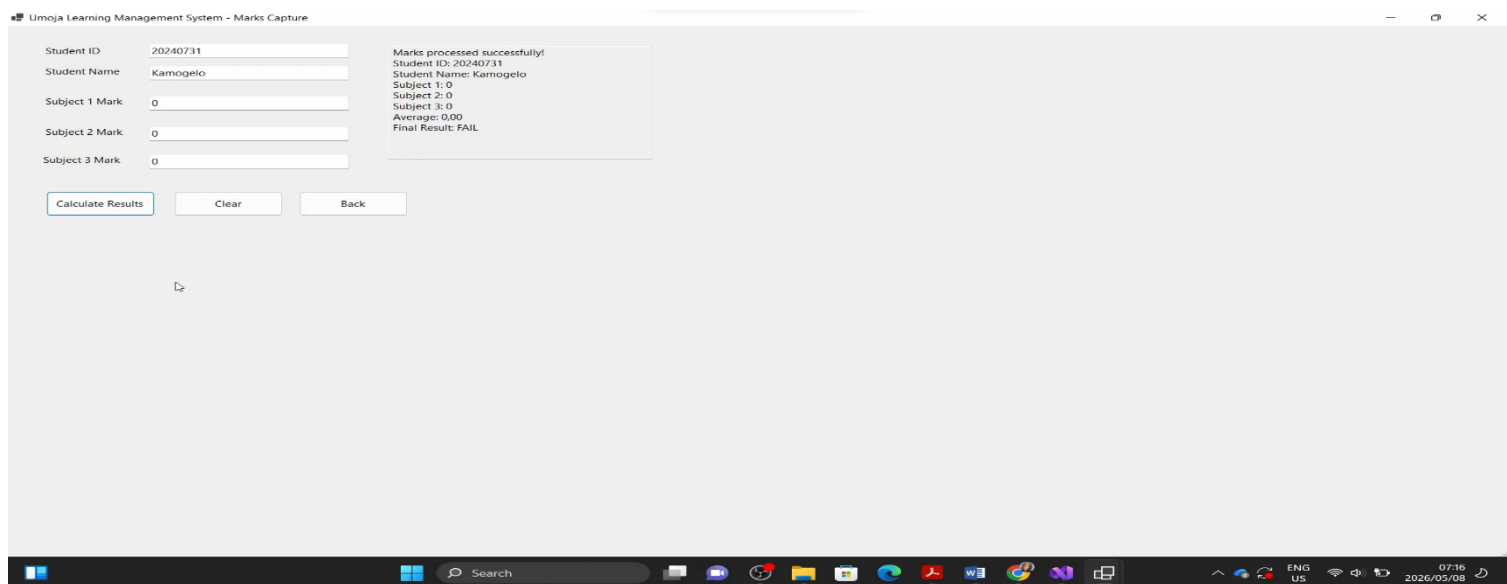
Umoja Learning Management System - Marks Capture

Student ID	20240731	Marks processed successfully! Student ID: 20240731 Student Name: Kamogelo Subject 1: 100 Subject 2: 100 Subject 3: 100 Average: 100.00 Final Result: PASS
Student Name	Kamogelo	
Subject 1 Mark	100	
Subject 2 Mark	100	
Subject 3 Mark	100	



Umoja Learning Management System - Marks Capture

Student ID	20240731	Marks processed successfully! Student ID: 20240731 Student Name: Kamogelo Subject 1: 0 Subject 2: 0 Subject 3: 0 Average: 0.00 Final Result: FAIL
Student Name	Kamogelo	
Subject 1 Mark	0	
Subject 2 Mark	0	
Subject 3 Mark	0	



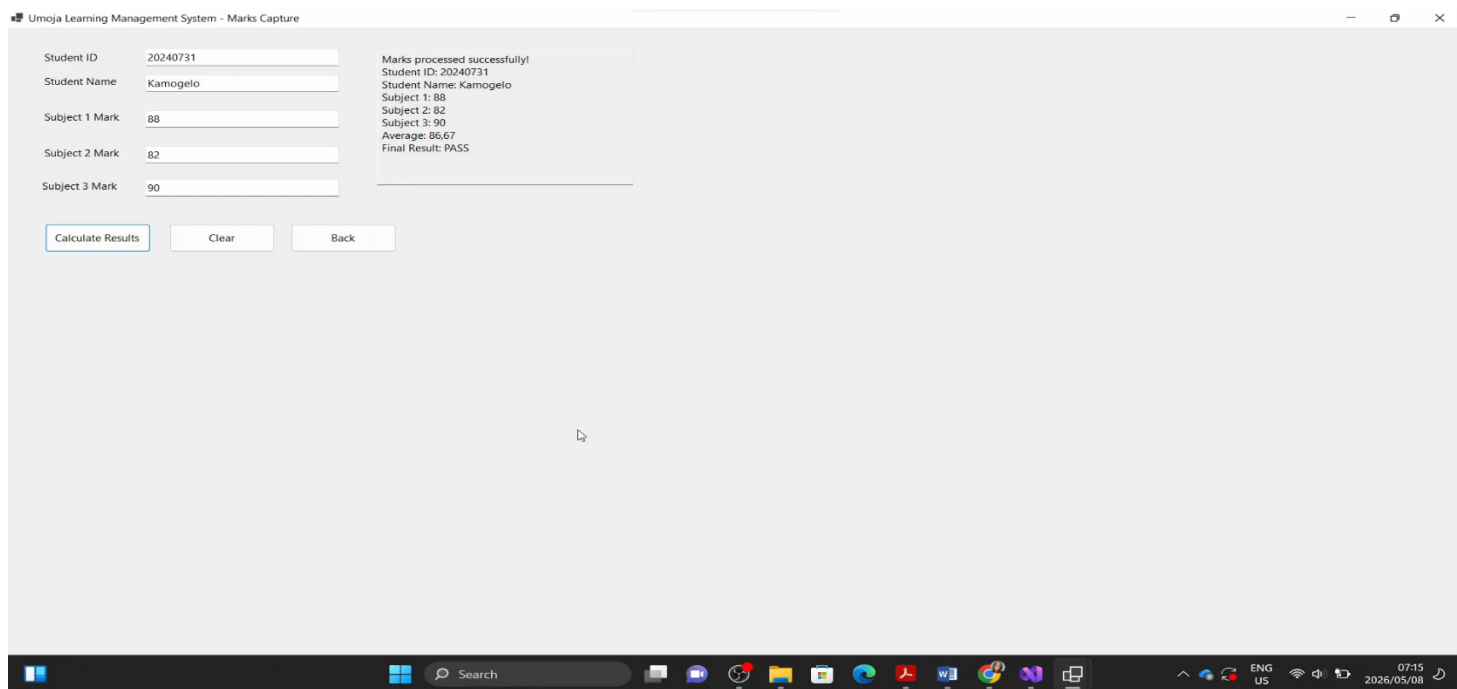
Method:

Repeated key operations 10 times to check consistency and stability.

Tests Performed:

Repeated login/logout cycle

Multiple student registrations



Multiple marks calculations

Report generation multiple times

Results:

All operations produced consistent results with no crashes after fixes. The application remained stable during repeated use.

Findings:

The system is reliable for its current scope.

Overall Non-Functional Testing Summary

The ULMS application performs as required after the implemented fixes. The main areas of improvement were Performance and Input validation. The system is now suitable for basic internal use.

Task 5: Debugging source code

The following are the source code defects/bugs I have identified.

Defect 1: Login Validation login

Module: FrmLogin.cs

Bug code

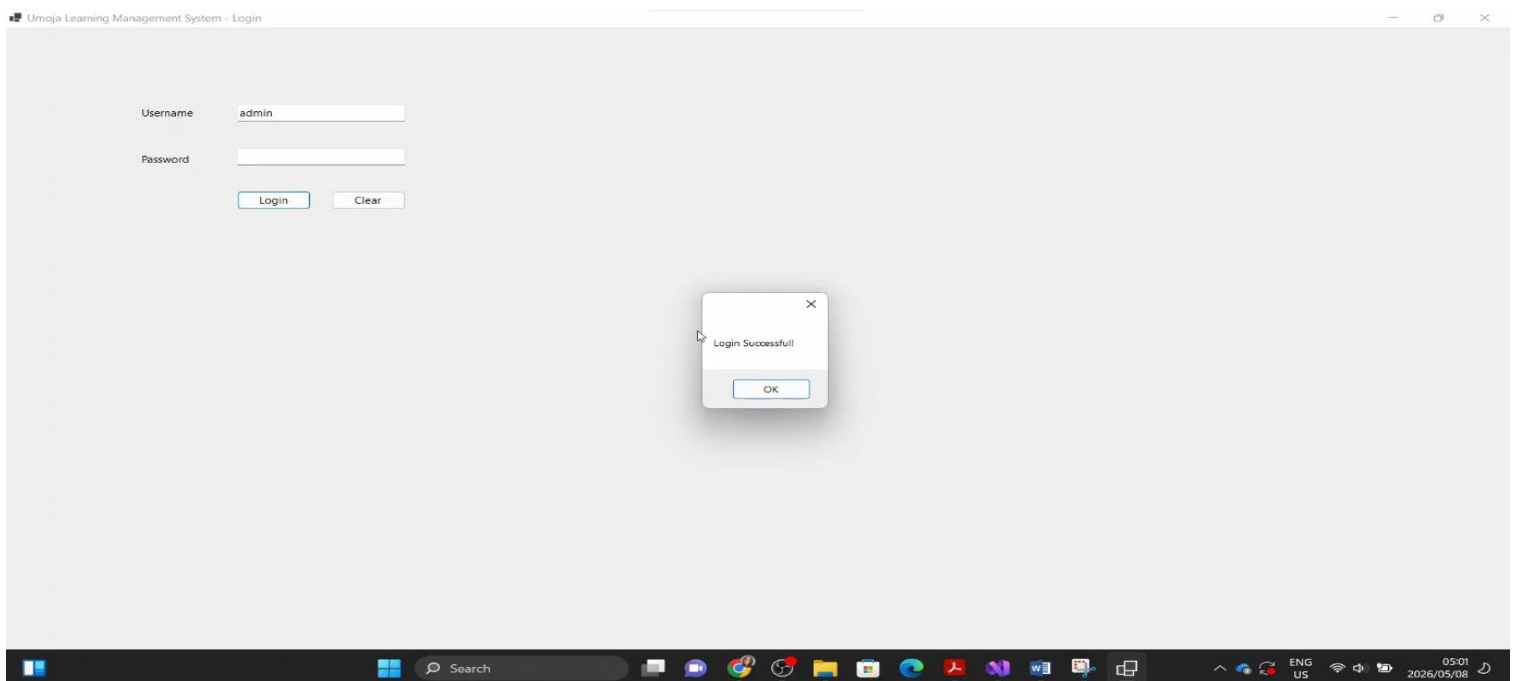
```
// Intentional faulty validation logic (for testing scenario)  
if (username == "admin" || password == "1234")
```

The login system allowed incorrect access because it used logical OR instead of logical AND. This means that the condition becomes true if either the username OR the password is correct instead of requiring both.

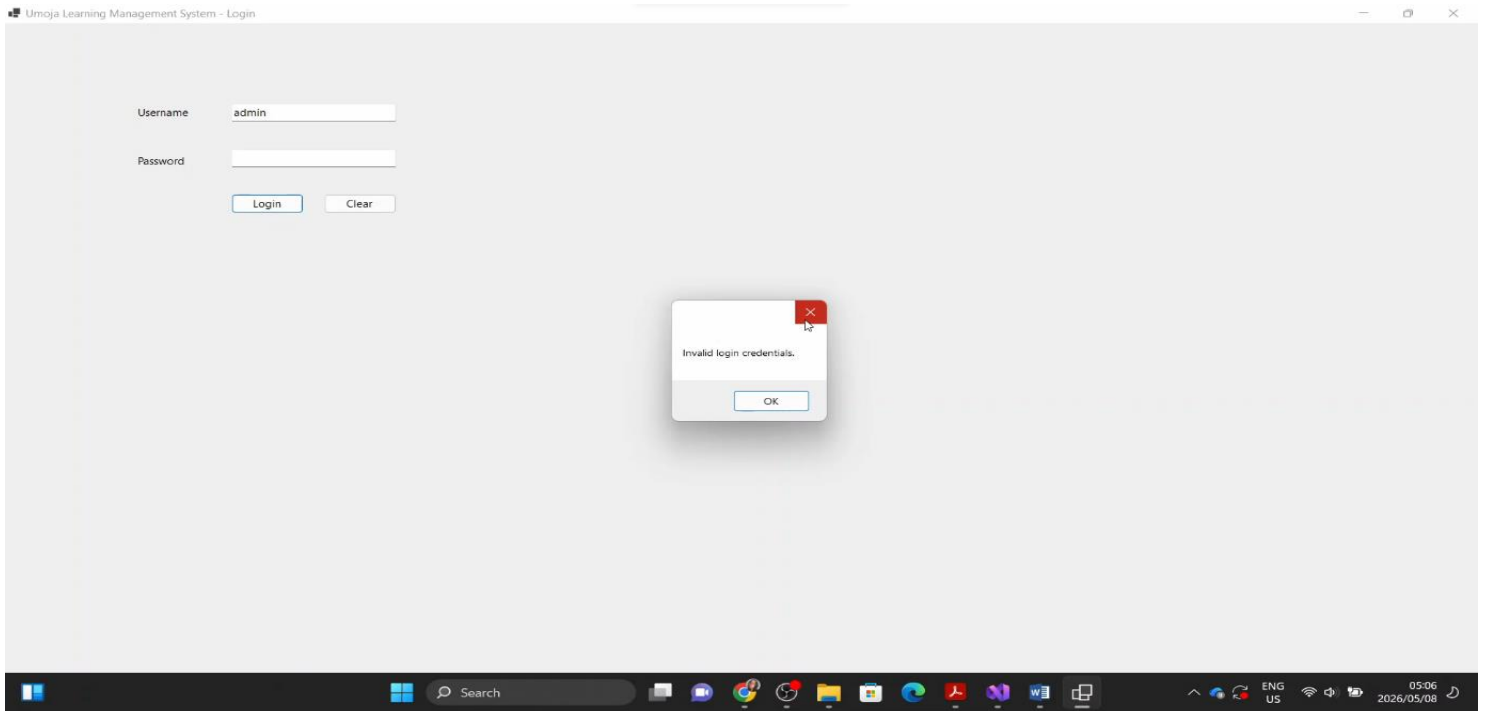
Corrected code

```
if (username == "admin" && password == "1234")
```

BEFORE FIX



AFTER FIX



Defect 2: Average Calculation Error

Module: FrmMarksCapture.cs

Bug code

```
// Intentional faulty calculation  
record.Average = record.Subject1 + record.Subject2 + record.Subject3 / 3;  
  
"Average: " + record.Average + Environment.NewLine +
```

Student average marks were calculated incorrectly. Due to order of operation in C#, only Subject3 was divided by 3 before adding the other subjects. Average output was not displayed in 2 decimal places.

Corrected code

```
record.Average = (record.Subject1 + record.Subject2 + record.Subject3) / 3;  
  
"Average: " + record.Average.ToString("F2") + Environment.NewLine +
```

BEFORE FIX

Umoja Learning Management System - Marks Capture

Student ID	20240731	Marks processed successfully! Student ID: 20240731 Student Name: Kamogelo Subject 1: 88 Subject 2: 82 Subject 3: 90 Average: 200 Final Result: PASS
Student Name	Kamogelo	
Subject 1 Mark	88	
Subject 2 Mark	82	
Subject 3 Mark	90	

Umoja Learning Management System - Marks Capture

Student ID	20240731	Marks processed successfully! Student ID: 20240731 Student Name: Kamogelo Subject 1: 88 Subject 2: 82 Subject 3: 90 Average: 86.66666666666667 Final Result: PASS
Student Name	Kamogelo	
Subject 1 Mark	88	
Subject 2 Mark	82	
Subject 3 Mark	90	

AFTER FIX

Umoja Learning Management System - Marks Capture

Student ID	20240731	Marks processed successfully! Student ID: 20240731 Student Name: Kamogelo Subject 1: 88 Subject 2: 82 Subject 3: 90 Average: 86.67 Final Result: PASS
Student Name	Kamogelo	
Subject 1 Mark	88	
Subject 2 Mark	82	
Subject 3 Mark	90	

Defect 3: Missing Input Validation

Module: FrmMarksCapture.cs

Bug code

```
// Intentional weak validation and faulty average logic for testing purposes
MarkRecord record = new MarkRecord();

record.StudentId = txtMarkStudentId.Text;
record.StudentName = txtMarkStudentName.Text;
record.Subject1 = Convert.ToDouble(txtSubject1.Text);
record.Subject2 = Convert.ToDouble(txtSubject2.Text);
record.Subject3 = Convert.ToDouble(txtSubject3.Text);
```

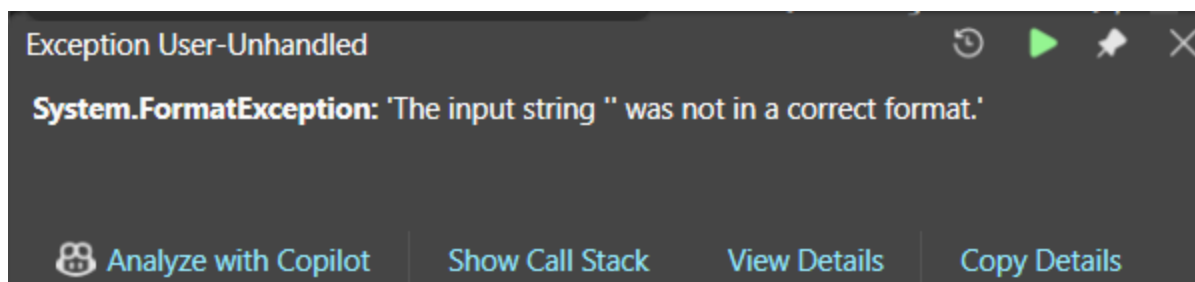
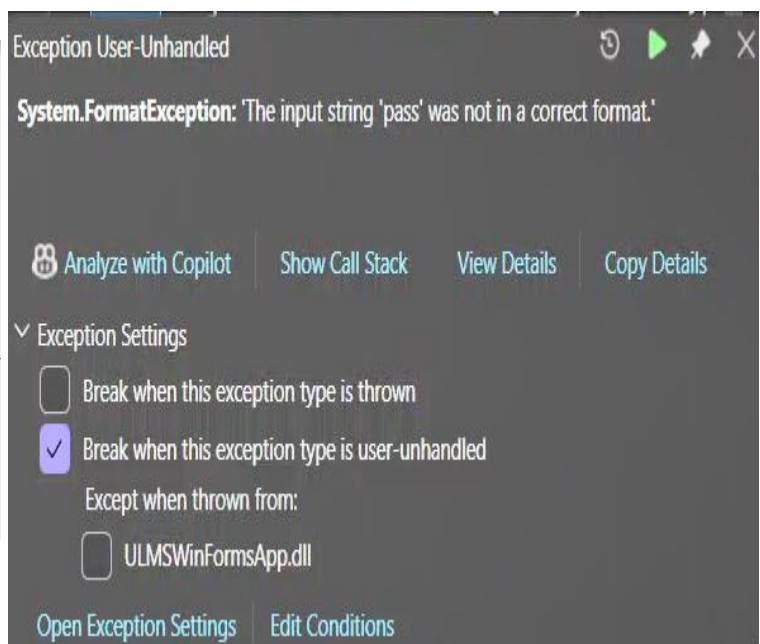
No validation for empty fields or non-numeric input in marks capture.

Corrected code

```
if (string.IsNullOrEmpty(txtSubject1.Text) || string.IsNullOrEmpty(txtSubject2.Text) || string.IsNullOrEmpty(txtSubject3.Text))
{
    MessageBox.Show("Please enter marks for all subjects.");
    return;
}
try
{
    record.Subject1 = Convert.ToDouble(txtSubject1.Text);
    record.Subject2 = Convert.ToDouble(txtSubject2.Text);
    record.Subject3 = Convert.ToDouble(txtSubject3.Text);
}
catch
{
    MessageBox.Show("Please enter valid numeric marks.");
    return;
}
```

BEFORE FIX

Student ID	20240731
Student Name	Kamogelo
Subject 1 Mark	pass
Subject 2 Mark	fail
Subject 3 Mark	pass



AFTER FIX

Umoja Learning Management System - Marks Capture

Student ID:

Student Name:

Subject 1 Mark:

Subject 2 Mark:

Subject 3 Mark:

Please enter marks for all subjects.

Umoja Learning Management System - Marks Capture

Student ID:

Student Name:

Subject 1 Mark:

Subject 2 Mark:

Subject 3 Mark:

Please enter valid numeric marks.

Defect 4: Performance Issue in Reports

Module: FrmReports.cs

Bug code

```
string reportType = cmbReportType.Text;
string studentId = txtReportStudentId.Text;

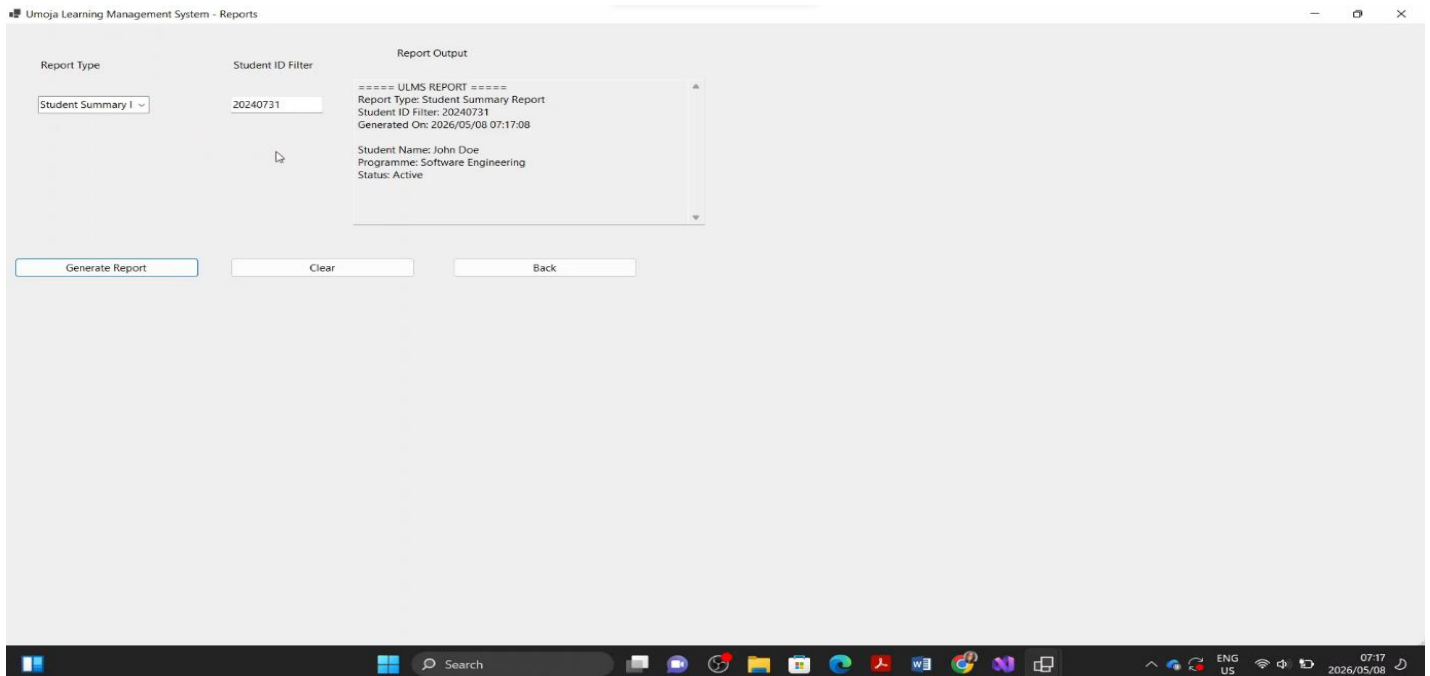
Thread.Sleep(4000);
```

Artificial delay of 4 seconds when generating reports cause the generator to display only after that.

Corrected code

```
Thread.Sleep(700);
```

Artificial delay was dropped to 0.7 seconds when generating reports for better user experience.



Defect 5: Missing Input Validation in report generator.

Module: FrmReports.cs

Bug code

```
string reportType = cmbReportType.Text;
string studentId = txtReportStudentId.Text;

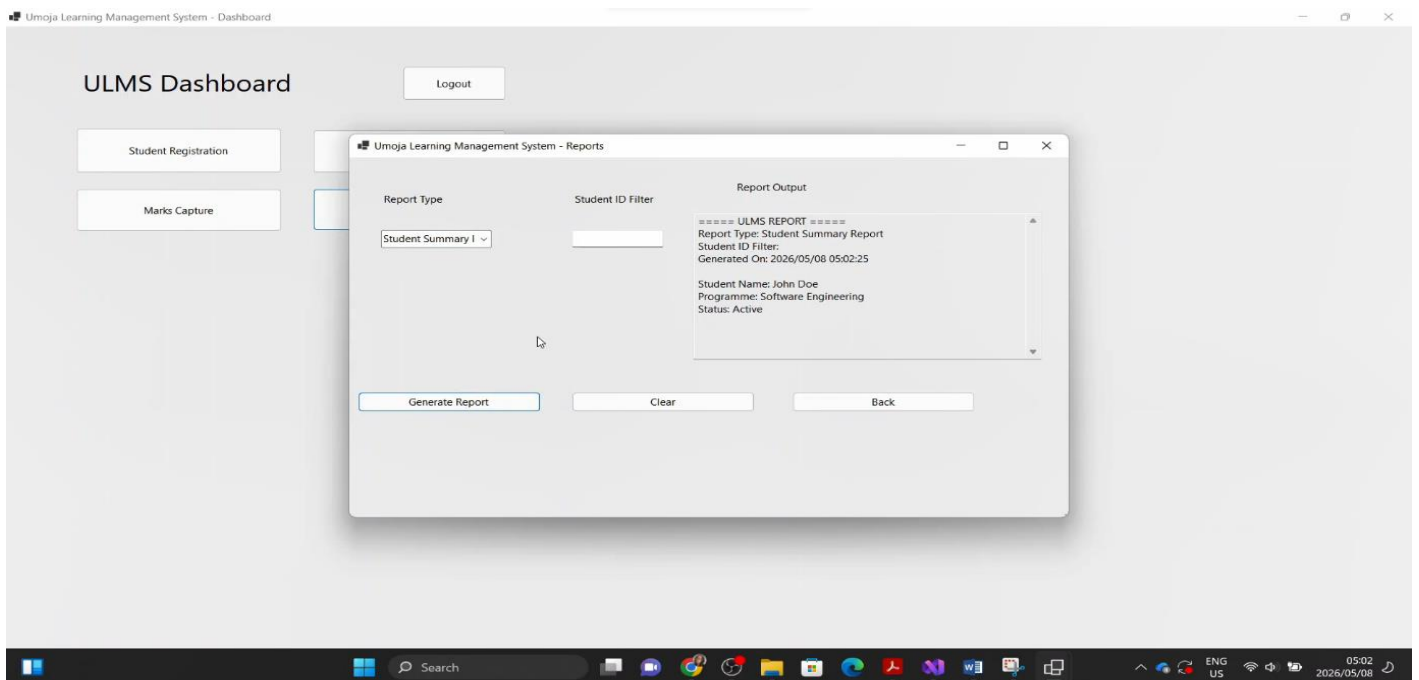
Thread.Sleep(4000);
```

No validation for empty Student ID

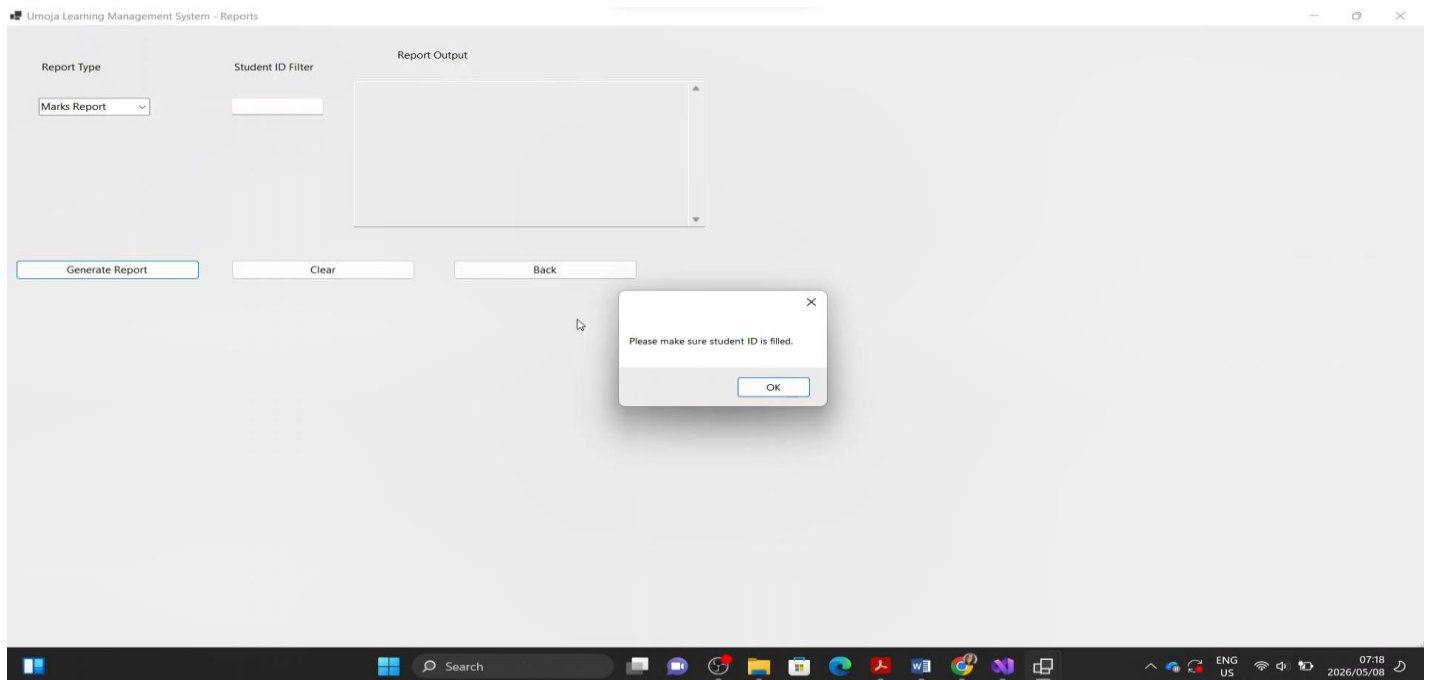
Corrected Code

```
string reportType = cmbReportType.Text;  
string studentId = txtReportStudentId.Text;  
  
if(string.IsNullOrEmpty(studentId))  
{  
    MessageBox.Show("Please make sure student ID is filled. ");  
    return;  
}
```

BEFORE FIX



AFTER FIX



Defect 6: Missing Input Validation and Exception Handling in Student Registration

Module: FrmStudentRegistration.cs

Bug code

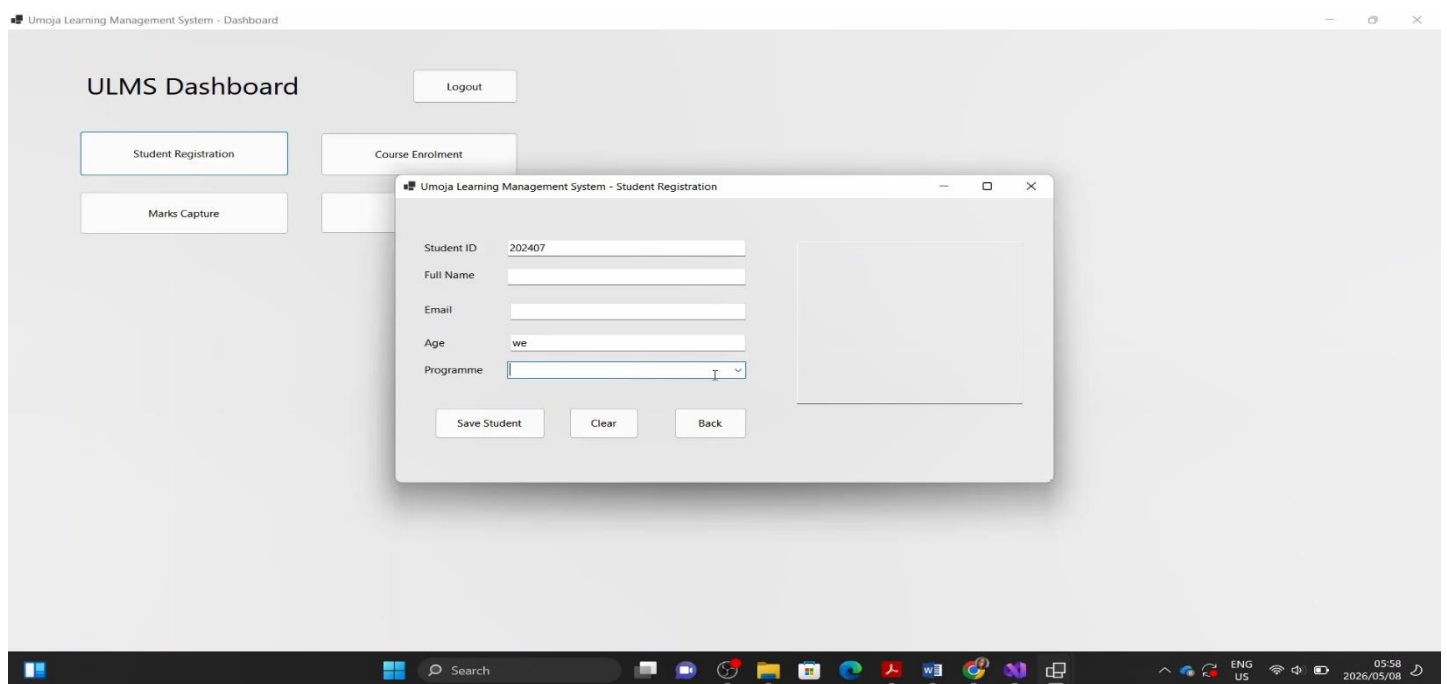
```
// Intentional weak validation for testing purposes
Student student = new Student
{
    StudentId = txtStudentId.Text,
    FullName = txtFullName.Text,
    Email = txtEmail.Text,
    Age = int.Parse(txtAge.Text),
    Programme = cmbProgramme.Text
};
```

Application crashes if non-numeric value is entered in Age field or if fields are left empty. There was a direct use of `int.Parse()` without validation or exception handling.

Corrected Code

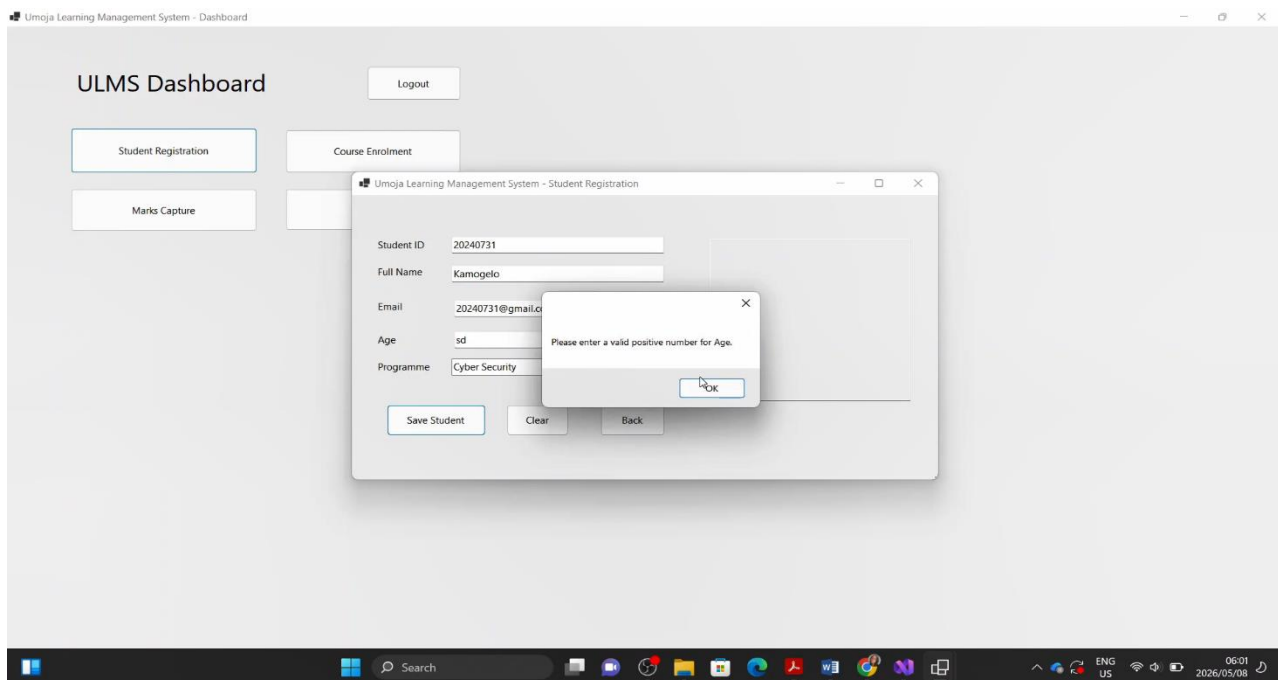
```
if (string.IsNullOrWhiteSpace(txtStudentId.Text) ||  
    string.IsNullOrWhiteSpace(txtFullName.Text) ||  
    string.IsNullOrWhiteSpace(txtEmail.Text) || string.IsNullOrWhiteSpace(txtAge.Text) || cmbProgramme.SelectedIndex == -1)  
{  
    MessageBox.Show("Fields should not be empty.");  
    return;  
}  
if (!int.TryParse(txtAge.Text, out int age) || age <= 18)  
{  
    MessageBox.Show("Please enter a valid age greater than 18.");  
    return;  
}  
Student student = new Student  
{  
    StudentId = txtStudentId.Text,  
    FullName = txtFullName.Text,  
    Email = txtEmail.Text,  
    Age = age,  
    Programme = cmbProgramme.Text  
};
```

BEFORE FIX





AFTER FIX



Defect 7: Missing Input Validation in course enrolment

Module: FrmCourseEnrollment.cs

Bug code

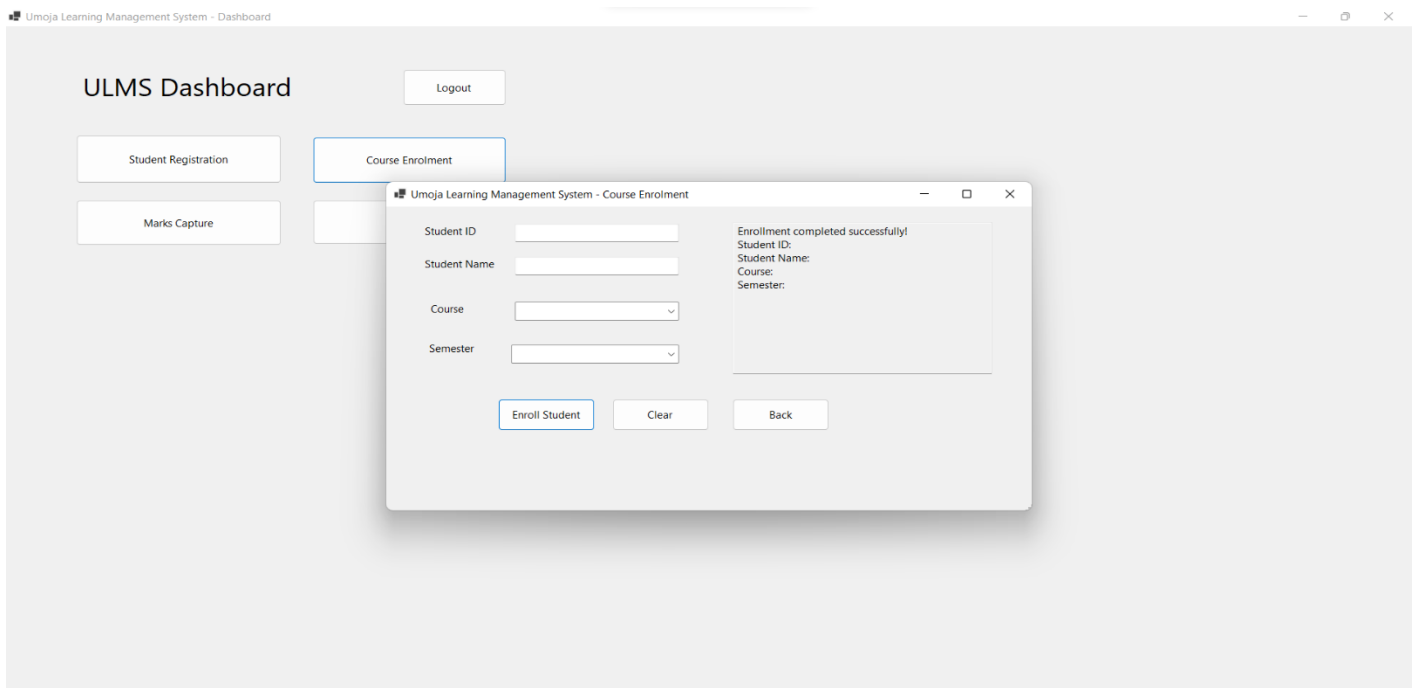
```
// Intentional weak business-rule validation for testing purposes
Enrollment enrollment = new Enrollment
{
    StudentId = txtEnrollStudentId.Text,
    StudentName = txtEnrollStudentName.Text,
    CourseName = cmbCourse.Text,
    Semester = cmbSemester.Text
};
```

Application still outputted and added course enrolment summary even if fields are left empty.

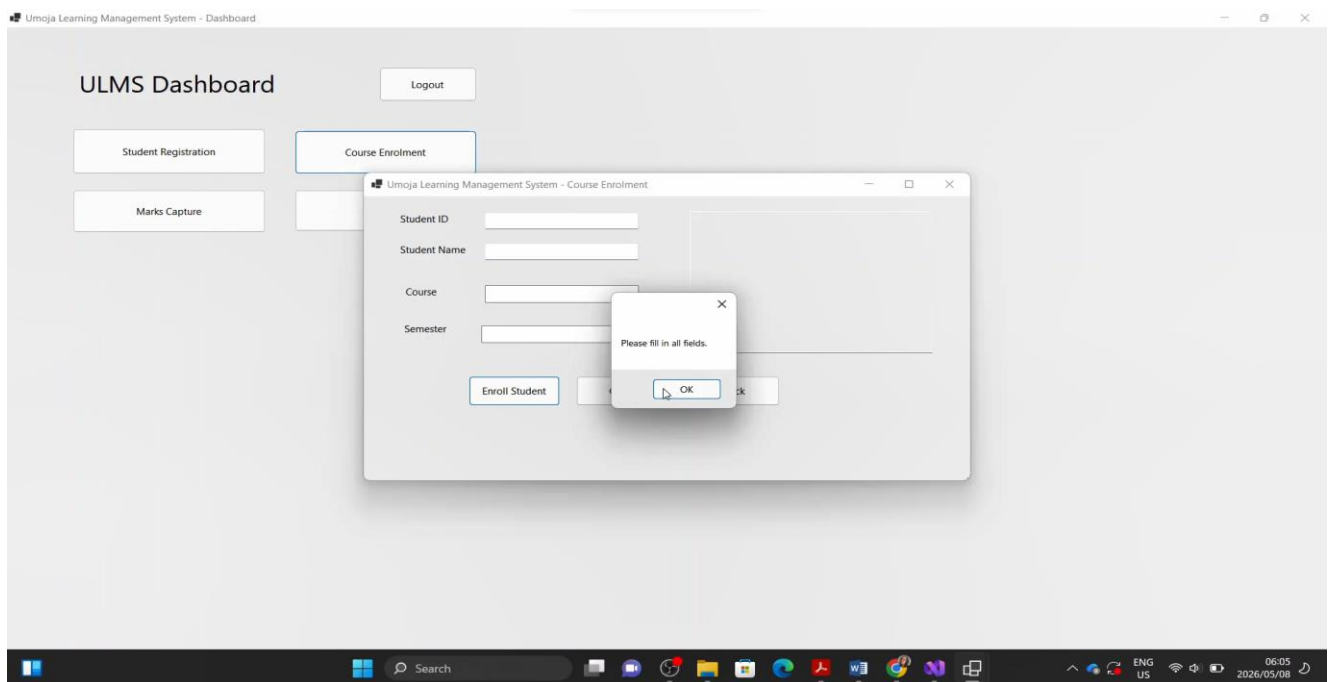
Corrected Code

```
if (string.IsNullOrEmpty(txtEnrollStudentId.Text) ||
    string.IsNullOrEmpty(txtEnrollStudentName.Text) ||
    cmbCourse.SelectedIndex == -1 || cmbSemester.SelectedIndex == -1)
{
    MessageBox.Show("Fields should not be empty");
    return;
}
Enrollment enrollment = new Enrollment
{
    StudentId = txtEnrollStudentId.Text,
    StudentName = txtEnrollStudentName.Text,
    CourseName = cmbCourse.Text,
    Semester = cmbSemester.Text
};
```

BEFORE FIX



AFTER FIX



TASK 6: Defect Reporting

Bug ID	Module	Description of Defect	Severity	Priority	Status	Resolution
BUG-001	FrmLogin.cs	Login validation used (OR) instead of && (AND)	High	High	Fixed	Replaced OR() with AND(&&)
BUG-002	FrmMarksCapture.cs	Wrong average calculation due to order or operator	High	High	Fixed	Added parentheses: (s1 + s2 + s3) / 3
BUG-003	FrmStudentRegistration.cs	No input validation causing it crashes on non-numeric Age	Medium	High	Fixed	Added TryParse + empty field checks
BUG-004	FrmReports.cs	Artificial delay (Thread.Sleep(4000)) causing slow performance	Medium	Medium	Fixed	Reduced delay to 700ms
BUG-005	FrmReports.cs	No validation for empty Report Type and Student ID	Medium	Medium	Fixed	Added proper validation checks
BUG-006	FrmMarksCapture.cs	Average displayed with many decimal places	Low	Low	Fixed	Changed to .ToString("F2") for 2 decimal places"
BUG-007	FrmCourseEnrolment.cs	No validation for empty fields	Medium	Medium	Fixed	Added proper validation checks
BUG-008	Overall System	Overall SystemLack of Data Persistence – Reports do not reflect actual registered students, marks, or enrolments (shows dummy/hardcoded data)	High	High	Open	Cannot be fixed – Requires database implementation

Declaration of authentication



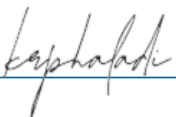
DECLARATION OF AUTHENTICITY

I Kamogelo Phaladi (FULL NAME)

hereby declare that the contents of this assessment SDT621 FA2

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

Signature: 

Date: 08/05/2026